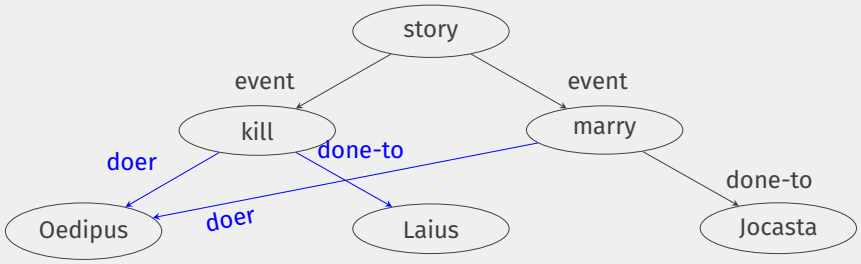
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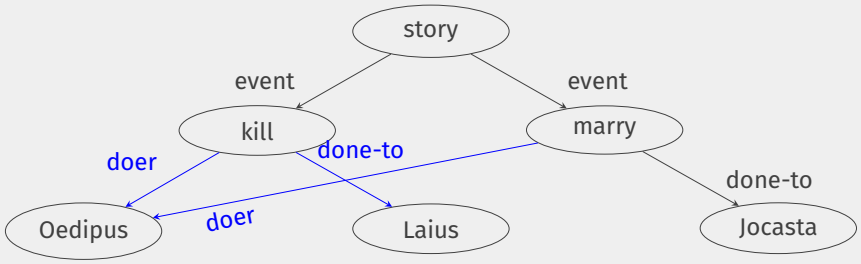
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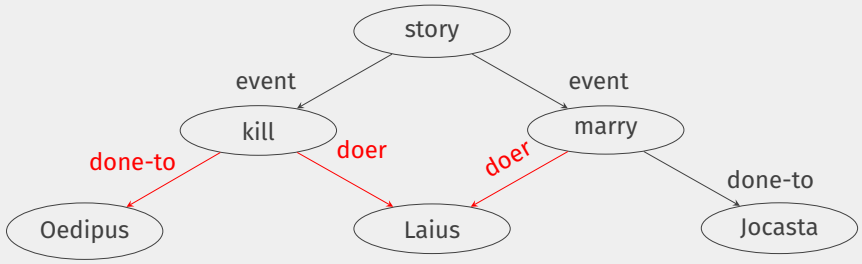


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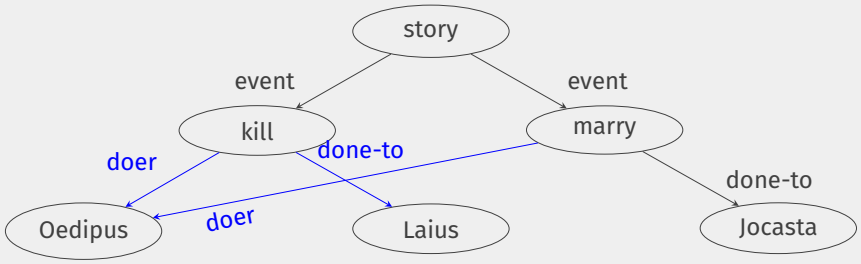


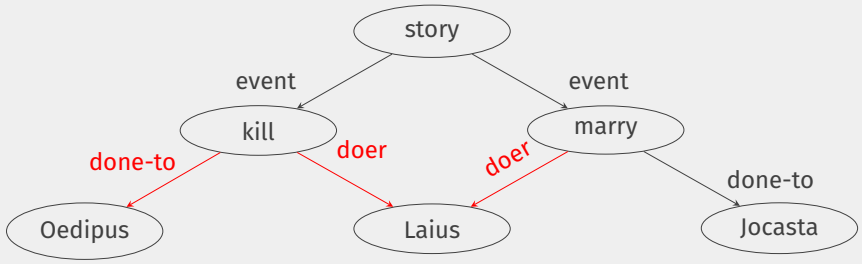


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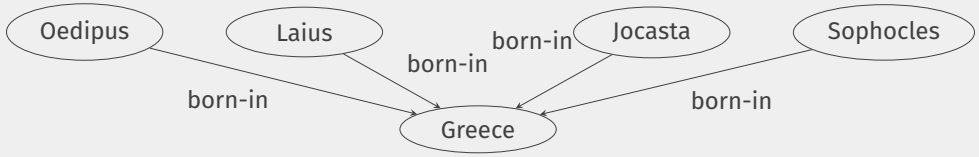
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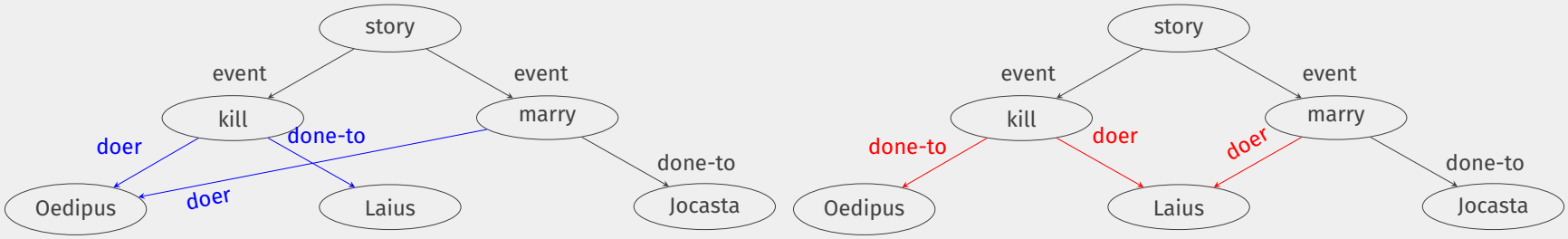
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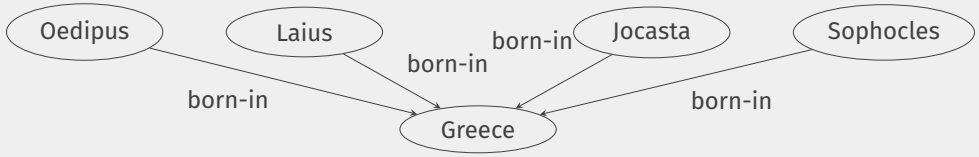
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How do we do it?

NLP1

How Language Works

Wilker Aziz



https://nlp-illc.github.io/nlp1/2026/

Fall 2026

Outline

1. How do we do it?
2. Mental Grammar
3. Mental Dictionary

How do we do it?

How do we do it?

> *In Sophocles' play, Oedipus killed his father, king Laius of Thebes, and married his mother, Jocasta.*

We know the thoughts that this sentence conveys because you and I

- have undergone an identical act of rote learning that links the sound of a word to its meaning,
- and, we know a code to translate between orders of words and combinations of thoughts.

The example is from [10].

The arbitrariness of the sign [Saussure]

The pairing of a sound with a meaning is wholly conventional

- we memorise the correspondence in childhood using something like a mental dictionary.

By fetching meaning from memory, using sound as a key, we are able to convey a concept from mind to mind virtually instantaneously.[2]

[2] For sign languages, substitute gesture for sound.

The arbitrariness of the sign [Saussure]

The pairing of a sound with a meaning is wholly conventional

- we memorise the correspondence in childhood using something like a mental dictionary.

By fetching meaning from memory, using sound as a key, we are able to convey a concept from mind to mind virtually instantaneously.[2]

These memorised items are what we call **listemes** [2].

- listemes: kick, the, bucket, sail, boat
- also a listeme: kick the bucket
- not a listeme: sailing boat

[2] For sign languages, substitute gesture for sound.

Listemes cannot be the whole story

Productivity:

- Verb inflection: English (4 forms), Portuguese (50-ish), Turkish (two million)
- Derivation: learn, *un*learn, learn*able*, *un*learn*able*, ...
- Compounding: grandmother, great-grandmother, counterfactual, counter-program, microwave
- Multiple: unmicrowaveability.

Word Machines

If all at our disposal were a large table in the brain, we would quickly run out of brain.

We must be equipped with a system of word formation that recombines memorised information—call it **morphology**.

Some of the evidence, again, comes from children. Across the world's languages (incl. sign languages), children

- can form the plural of words outside their experience, inflect new verbs for tense (and other grammatical paraphernalia);
- make equivalent generalisations (and over-generalisations) on schedule [3, 7];

Sentence Machines

If interrupted at random while producing a sentence, an English speaker can continue with, on average, any of 10 words.

Assume an adult is able to produce and comprehend sentences with 20 words. That brings us to 10^{20} sentences.

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We use a code to translate between orders of words and combinations of thoughts—call it **grammar** (or **syntax**).

A grammar is a discrete combinatorial system in which a finite number of discrete elements (words) is sampled, combined and permuted to create larger structures (phrases and sentences).

The properties of the result are quite distinct from those of their elements, while still predictable.

In a discrete combinatorial system like language there can be an unlimited number of distinct combinations with an infinite range of properties.

Summary

The way language works then is that each person's brain contains

- a lexicon of symbols and the concepts they stand for—**a mental dictionary**,
- and a system that combines words to convey relationships amongst concepts—**a mental grammar**.

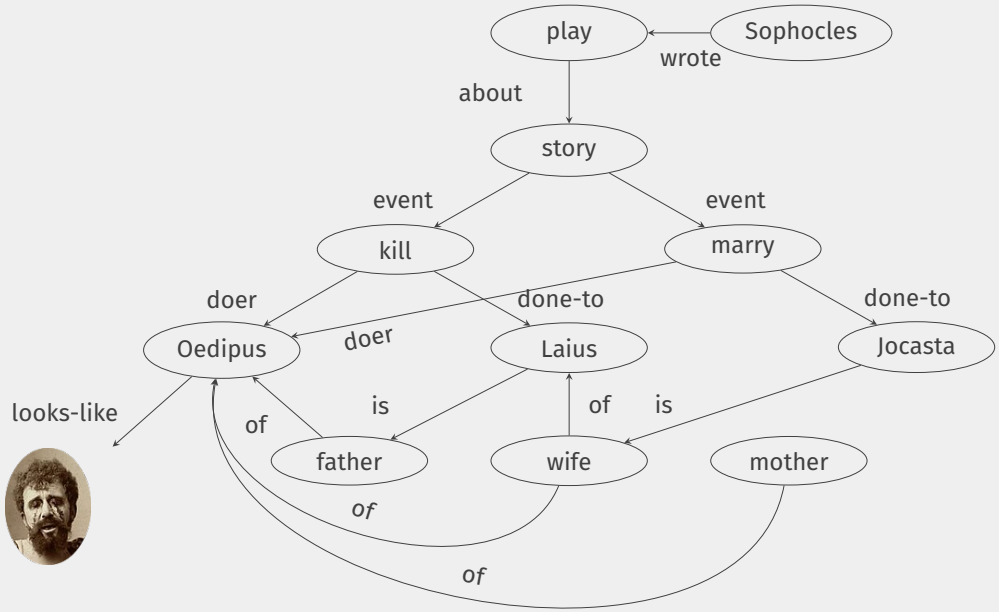
Next, we expand on these two modules:

- our goal is develop some appreciation for the complexity of the **data** we are about to embrace in this course.

Mental Grammar

Language conveys information

In Sophocles' play, Oedipus killed his father, king Laius of Thebes, and married his mother, Jocasta.



We know that

- Laius and Jocasta are Oedipus's parents
- Laius is the king of Thebes
- Oedipus did the killing and the marrying
- Laius is dead

because of the *order* of the words (and their meanings ofc).

Generative Grammar

The code we use is what we call a **generative grammar**.

Not to be confused with the use of *generative* in ML.

Not to be confused with *pedagogical grammars*

- Pedagogical grammars are prescriptive
- A linguist's generative grammar is descriptive

Grammaticality

To a linguist, grammaticality has nothing to do with conformity with a book of wisdom.

Rather, grammaticality is a hypothesis to be ascertained empirically.

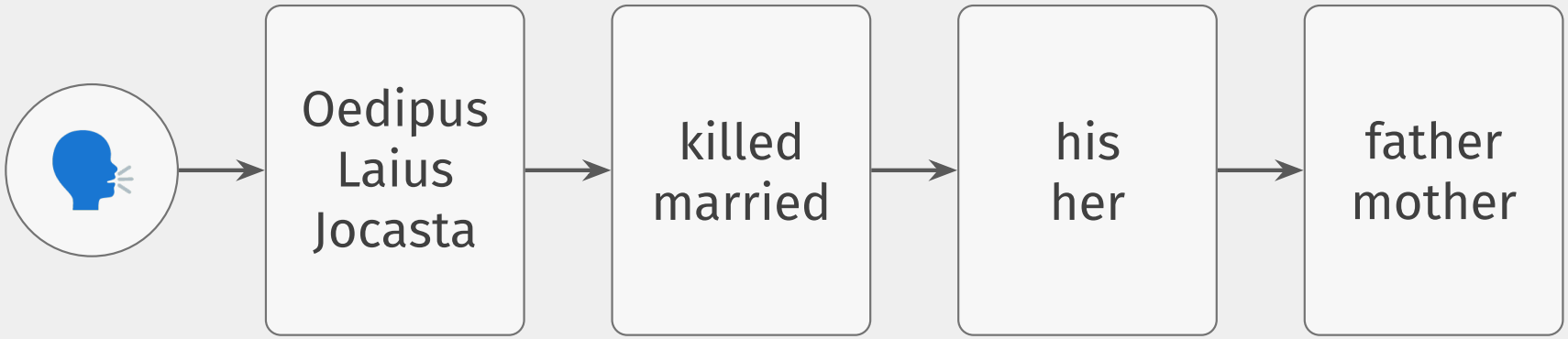
Something is ungrammatical if ordinary speakers have a ‘gut feeling’ that something is wrong:

- The child seems sleeping.
- Is raining.
- Sally poured the glass with water.
- This sentence no verb.
- This is not a complete. This either.

Combinatorial System

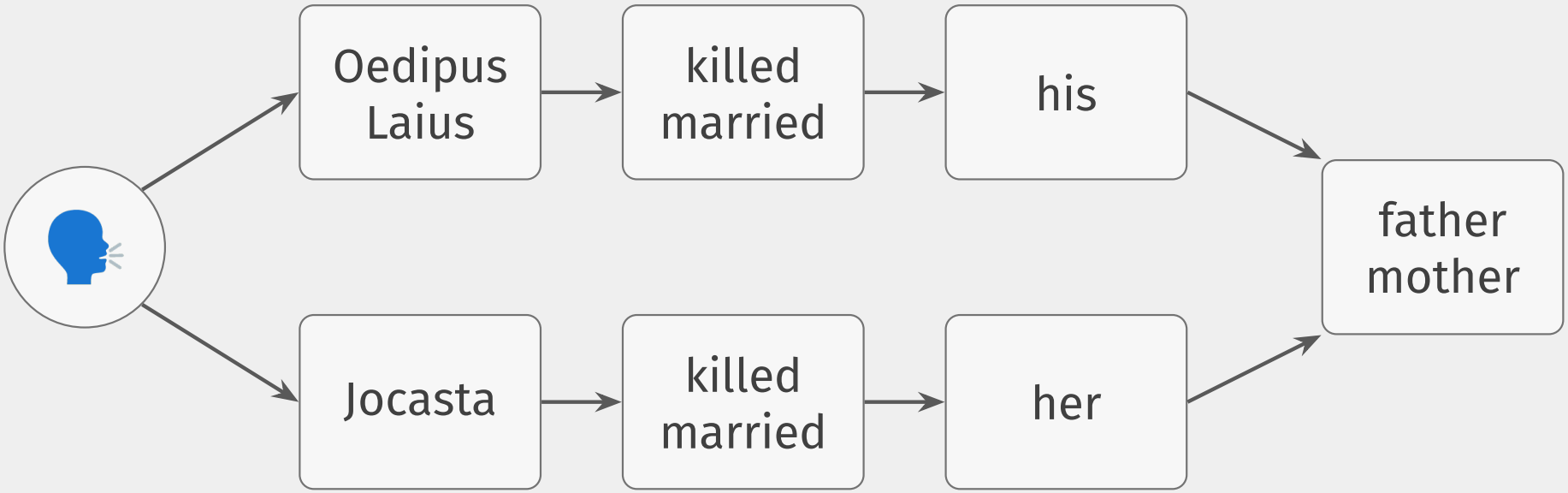
The principle underlying grammar is that of a combinatorial system: discrete elements are sampled, combined and permuted to create larger structures.

The simplest such system is a chain of word lists (formally, a finite-state automaton or FSA):



Unreasonable Redundancy

Word lists offer only rather crude ways to incorporate ‘dependencies’. For example, how the choice of pronoun in 3rd position depends on the system’s choice for the 1st position.



By the way,

These systems, albeit augmented with probabilities for its transitions, powered NLP tools for text completion, summarisation, translation, etc.

We will get back to these **Markov models** next week.

Universality of Complex Languages

But every one of the world's languages exhibit features that are well beyond FSAs:[3]

> *How$_1$ Ann Salisbury can claim that Pam Dawber's anger$_2$ at not receiving her fair share of acclaim for Mork and Mindy's success derives$_2$ from a fragile ego escapes$_1$ me.*

- long-range dependencies (e.g., How ... escapes me and Pam Dawber's anger... derives);
- embedded phrases (e.g., the amazing Ann Salisbury, who had been silent for months and now finally speaks up, can claim that)

[3] Recall that Everett's claims about Pirahã are not widely accepted; see my recent Ed post.

FSAs are Fundamentally Wrong

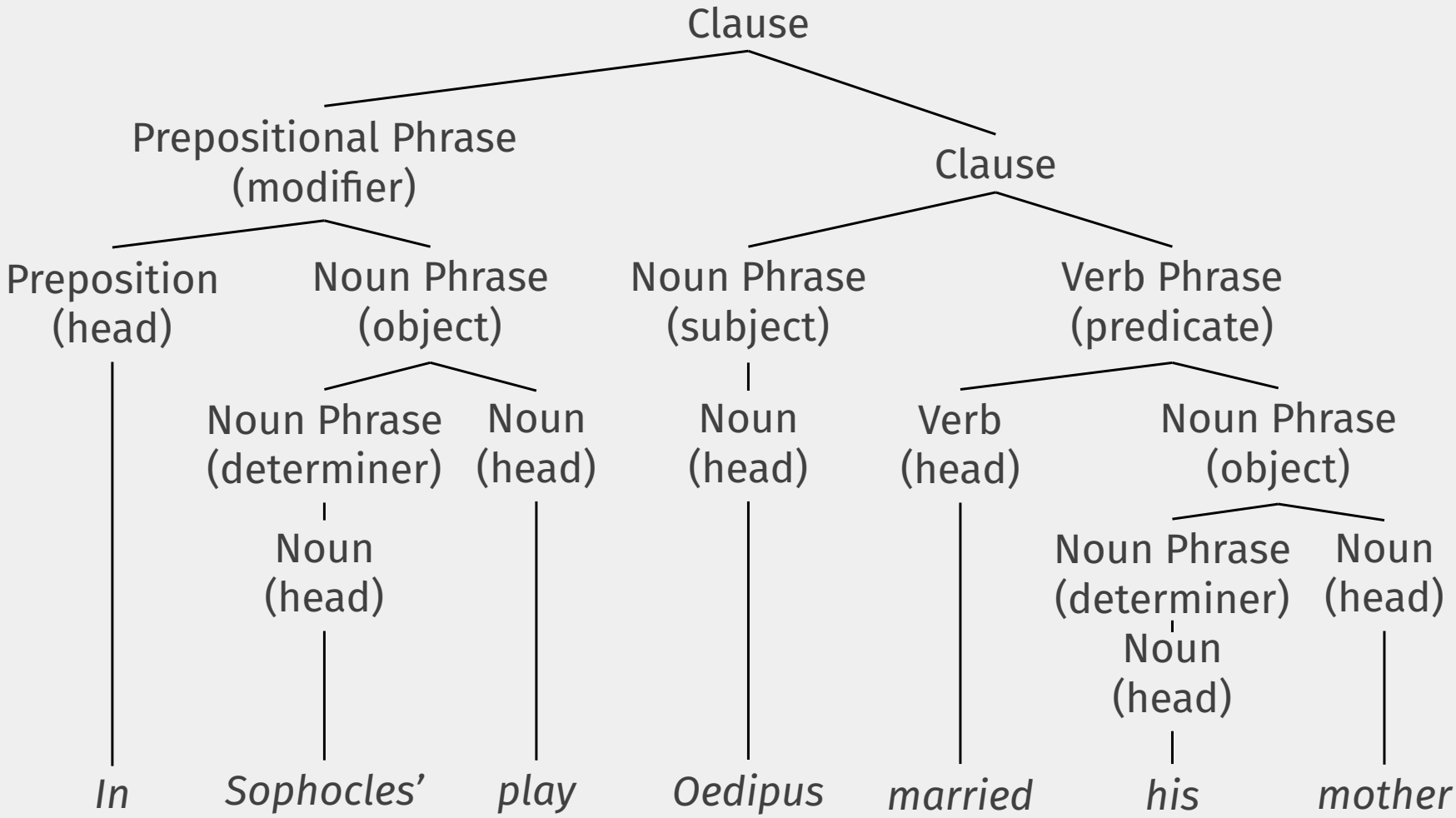
Without implausible degrees of redundancy, an FSA cannot handle such features of natural languages [6].

They would grow exponentially large, but our brains and learning time is the same whether we are to learn long-range dependencies or not.

And we all do!

Modularity

A device that comes close to meeting the needs of all of the world's languages is a tree.



Discoveries: noun phrases

When a linguist suggests that a noun phrase can have the form

$$NP \rightarrow (det)\ A^*\ N$$

This a testable hypothesis, namely

> *An optional determiner, followed by any number of adjectives, followed by a noun function like a noun.*

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> *An optional determiner, followed by any number of adjectives, followed by a noun function like a noun.*

- *the little, happy dog* is a *dog*
- *Sophocle's play* is a *play*
- *governor of California* is a *governor*
- *his mother* is a *mother*

The word that dictates how the phrase functions is the pharase's **head**. The head may impose strong constraints on the kinds of role players (e.g., *of California*), modifiers (e.g., *little, happy*) and specifiers (e.g., *Sophocle's*) that are welcome/necessary.

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This a testable hypothesis, namely.

> *A verb followed by a noun phrase functions like a verb.*

- *married his mother*
- *married his mother on Tuesday*
- *married his mother over the objections of his girlfriend*

These all function like the head *married*. The head also imposes constraints on the kinds of role players (like *his mother*) and modifiers (like *on Tuesday*) that are welcome/necessary.

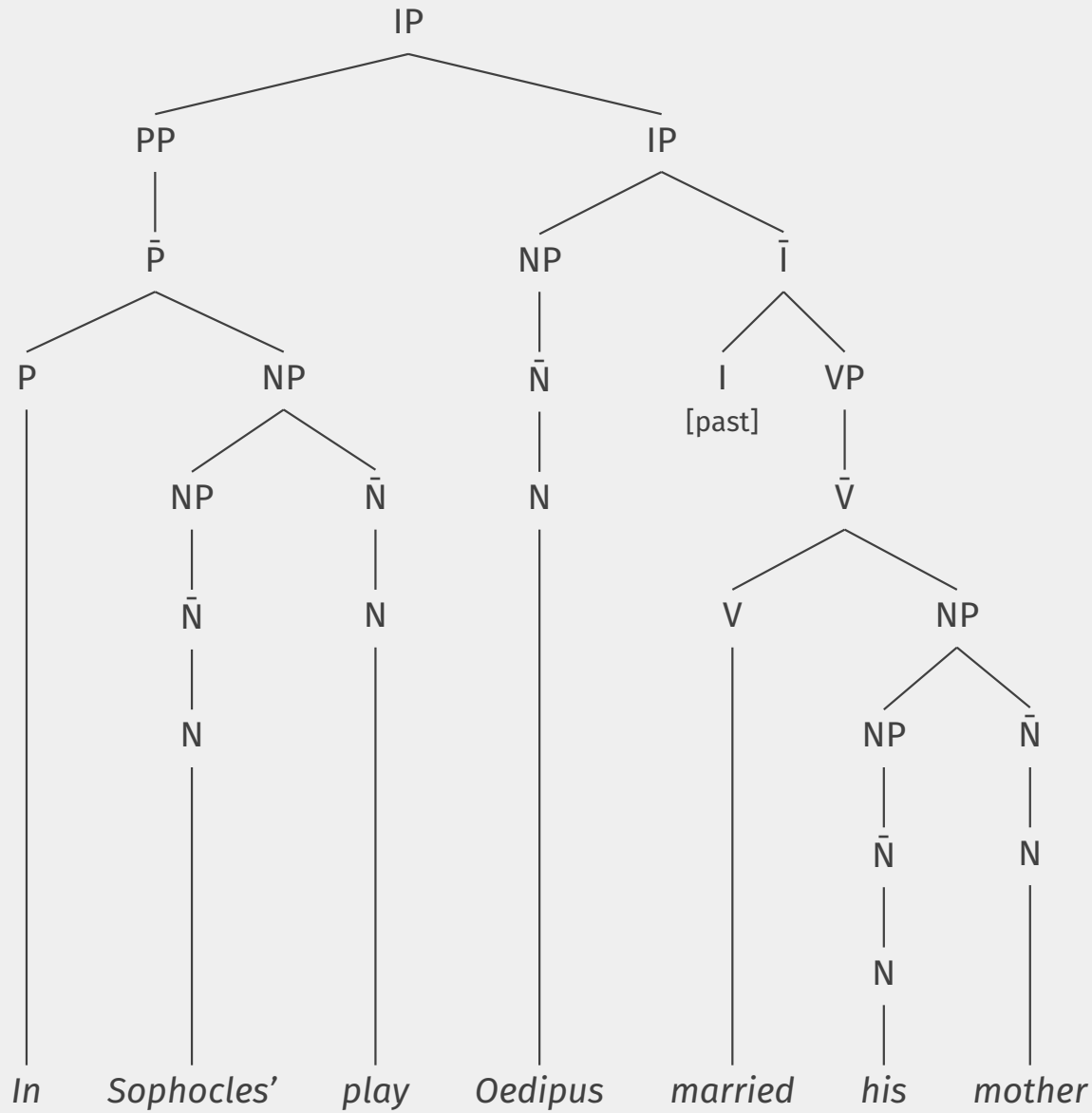
There is in fact a common design to phrases:

1. XP → (SPEC) $\bar{X}$ YP*
   An X-phrase consists of an optional subject (specifier), followed by an X-bar, followed by any number of modifiers.

2. $\bar{X}$ → X ZP*
   An X-bar consists of a head word, followed by any number of role-players.

A phrase inherits the properties of its head. These properties include all kinds of things (meaning, selectional preferences, even pronunciation) which are stored in the mental dictionary.

An X-bar analysis of our running example



X-bar beyond English

X-bar holds quite solidly across languages, with complications I omitted here.

For example,

- the trees are in fact mobiles because the order (left-to-right or right-to-left) is a language-specific parameter
- more of this kind of inaudible (but empirically testable) structure

By the way,

Tree-based models were very common in pre-2010 NLP.

But more often than not, because of difficulties with probabilistic modelling and statistical learning, even though we often relied on tree formalisms we did not always rely on the linguistic theories in which these formalisms were developed.

The textbook has a couple of historical chapters, if you are interested (optionally, check Chapters 19 and 20).

I will try to share an optional reading list with some exciting, modern resurgences of syntax in NLP. In the meantime, here's someone whose work you can check
`https://stanojevic.github.io`

Not Even the Surface

This is really just to get you interested, and to give you a sense of what the logic that organises our mental life might look like.

A complete theory needs to account for so much more:

- inaudible traces, selectional preferences, etc;
- emphasis on biological plausibility (evolvability, feasibility of computation and learning, etc).

Some of these theories inspired a lot of research in NLP.

Today's large-scale NLP systems mostly circumvent an explicit account of the underlying logic of language. But there is a lot of evidence that their internal functions and representations are coherent with much of what we know about grammar.[4]

[4] See for example [12, 4].

Mental Dictionary

Back to the Word Machine

In Kivunjo, **Näïkìḿlyïià** conveys ‘he is eating it for her’

- N-: the word is the focus of that point of the conversation.
- -ä-: subject agreement marker, identifies the eater as falling into Class 1 of the sixteen gender classes.
- -ï-: present tense.
- -kì-: object agreement marker, identifies that the thing eaten falls into gender Class 7.
- -ḿ-: a benefactive marker, indicating for whose benefit the action is taking place, a member of gender Class 1.
- -lyì-: the verb ‘to eat’.
- -ī-: an applicative marker, indicating that the verb’s cast of players has been augmented by one additional role, in this case the benefactive.
- -à: a final vowel, which can indicate indicative versus subjunctive mood.

About half a million forms, for this one verb *-ilì*! Something remarkable must be behind this.

Morphology

Like sentences, words seem to come out of a combinatorial system, but with its own principles.

While grammar combines words into phrases and sentences, **morphology** combines **morphemes** into words.

Even though the systems have some similarities, the are good reasons to regard them as (at least) two.[5]

[5] An option that's of textbook quality but more accessible to someone who is not in a linguistics track is [9].

Morphemes, Listemes and Words

The mental dictionary is a metaphor for two things, really.

A repository of rote-memorised items and all the conventional properties that you could only have learnt that way. These are **listemes** and **morphemes**.[6]

And the combinatorial system—**morphology**—that combines these component parts.

[6] You can think of listemes being a larger set than morphemes; the difference is not too important for us.

Example: morpheme

Consider something like -ed in English, we need information of the kind:[7]

- sound: *d*
- part of speech: suffix
- use 1: past tense of a verb
- use 2: perfect participle of a verb
- use 3: passive participle of a verb
- use 4: adjective formed from a verb

The point is not that the mind has these little records for our contemplation. The point is that it must have something that amounts to the same effect of having it (or something like it).

[7] Perhaps these different uses could be regarded as different items.

Example: listeme

Oedipus

- sound: [ˈiːdɪpəs]
- part of speech: noun

It is not clear that relational knowledge



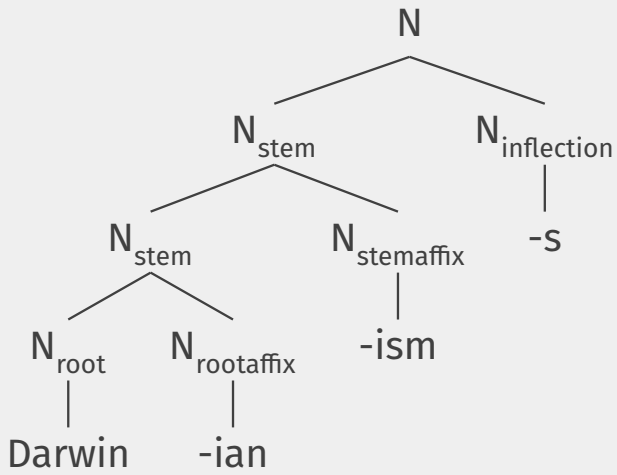
such as `looks-like`

needs to be stored here (it likely isn't; the mind is very modular!).

Example: morphological analysis

Word structure can be thought of as an extension of phrase structure. As we dissected a noun phrase as smaller noun-ish things, characterised by a head noun, when we get to the head, we continue to dissect the word into smaller noun-ish pieces.

- Inflection: N → Nstem Ninflection
  dog -s
- Compounds: N → Nstem Nstem
  sushi-lover
- Even stems decompose
  Darwin -ian -ism -s



How Large is the Mental Dictionary?

Let's focus on the notion of listeme: a chunk that needs to be memorised.[8]

Shakespeare's writings contain 15,000 listemes. How many do you think you know?

[8] Recall: 'kick', 'the', 'bucket', 'sail' and 'boat' are listemes and so is 'kick the bucket', but 'sailing boat' is not.

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A US-based estimate puts the figure at

- 13,000: average six-year-old
- 45,000: high-school graduates;

With such figures, preliterate kids must be acquiring a new word every two waking hours, day in, day out.

[8] Recall: 'kick', 'the', 'bucket', 'sail' and 'boat' are listemes and so is 'kick the bucket', but 'sailing boat' is not.

How Trivial is to Learn a Word?

You are the linguist, I am a native of a newly discovered tribe. A rabbit scurries by, and I shout **gavagai**.[9]

What does *gavagai* mean?

[9] This is the Gavagai problem, articulated by Quine [11].

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Children intuit the correct meaning of a word and avoid the vast number of logically impeccable, yet ridiculous, alternatives. Such a **scandal of induction** requires strong learning biases [1, 8, 5].

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Summary

The world of words is as amazing as the world of sentences.

From

- the sheer volume of memorised listemes
- to the sophistication of the combinatorial system
- to the efficiency with which it is tuned to a specific language

morphology is one of those wonders of natural design *which most justly excites our admiration*.

Morphology and insights from morphology have been and still are impactful in NLP (esp in relation to data sparsity and under-resourced languages). In T1, you will learn about the practical importance of word segmentation.

Orientation

Linguistics and Psycholinguistics concern how we acquire and use language and what our mental processes must look like.

For a while, their theories and findings were the main source of inspiration in NLP.

Initially, data was small and statistical models too simple. Prior knowledge, explicit structure and constraints offered a real edge.

Over time, data and compute grew fast, explicit structure and constraints became less appealing.

This course is *not* focused on the cognitive science of language. Now that we had this broad outline of what language is and how it works, we will focus on how **NLP models** work.

What Next?

Self-study

- Chapter 2 of the textbook (Words and Tokens);
- Before the start of module 2: tutorial 1 (sections Digital Text and Text Classification).

Deadlines

- Nov 2: register your groups and fill in the consent form for assessments.

Practicals

- T1 is packed with interesting stuff, don't miss out!
- Choose your task/dataset and get started with P1!

Next week: building blocks of NLU and NLG!

References I


Eve V. Clark.
The Lexicon in Acquisition.
Cambridge University Press, Cambridge, 1993.


Anna Maria Di Sciullo and Edwin Williams.
On the Definition of Word.
Number 14 in Linguistic Inquiry Monographs. MIT Press, Cambridge, MA, 1987.


Peter Gordon.
Level-ordering in lexical development.
Cognition, 21(2):73–93, 1985.


Jaap Jumelet.
Finding Structure in Language Models.
PhD thesis, University of Amsterdam, 2024.


Ellen M. Markman.
Categorization and Naming in Children: Problems of Induction.
MIT Press, Cambridge, MA, 1989.

References II



George A. Miller and Noam Chomsky.
Finitary models of language users.
In R. Duncan Luce, Robert R. Bush, and Eugene Galanter, editors, *Handbook of Mathematical Psychology*, volume 2, pages 419–491. John Wiley & Sons, New York, 1963.



Laura A. Petitto.
"language" in the prelinguistic child.
In Frank S. Kessel, editor, *The development of language and language researchers: Essays in honor of Roger Brown*, pages 187–221. Lawrence Erlbaum Associates, Inc., Hillsdale, NJ, 1988.



Steven Pinker.
Learnability and Cognition: The Acquisition of Argument Structure.
MIT Press, Cambridge, MA, 1989.

References III


Steven Pinker.
Words and Rules: The Ingredients of Language.
Basic Books, New York, 1999.


Steven Pinker.
The Sense of Style: The Thinking Person's Guide to Writing in the 21st Century.
Penguin Books, New York, 2014.


Willard Van Orman Quine.
Word and Object.
Technology Press of the Massachusetts Institute of Technology (joint publication John Wiley & Sons), Cambridge, MA, 1960.

References IV



Ian Tenney, Dipanjan Das, and Ellie Pavlick.
BERT rediscovers the classical NLP pipeline.
In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, Florence, Italy, July 2019. Association for Computational Linguistics.